

Student Performance Analytics System

Objective

The main objective of this project is to analyze student performance using marks and attendance data.

This project helps to understand how students are performing in different subjects and how attendance affects their marks. It also helps in identifying top-performing and low-performing students.

Dataset Description

The dataset contains information of 30 students with the following columns:

- **Student Name** – Name of the student
- **Math Marks** – Marks scored in Math
- **Science Marks** – Marks scored in Science
- **English Marks** – Marks scored in English
- **Attendance** – Attendance percentage of students

The dataset is clean and structured, and it is used for analysis and visualization.

Analysis

In this project, the following analysis is performed:

- Displayed first and last 5 records of dataset
- Checked dataset shape and column names
- Checked missing values and duplicate records
- Verified data types of each column

Basic Calculations:

- Calculated **average marks** of each subject
- Found **maximum and minimum marks**
- Calculated **standard deviation** to understand variation

Additional Analysis:

- Created a new column **Total Marks** (sum of all subjects)

- Identified **top-performing student**
 - Identified **lowest-performing student**
 - Found subject with **highest average marks**
 - Assigned **grades (A, B, C, D)** based on total marks
 - Ranked students based on performance
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Visualizations

The following graphs are created to understand the data better:

- **Bar Chart** → Shows average marks of each subject
- **Histogram** → Shows distribution of marks
- **Pie Chart** → Shows attendance distribution
- **Scatter Plot** → Shows relationship between attendance and total marks

These visualizations help in better understanding of student performance.

Insights

1. Students scored highest marks in **English subject**.
 2. Students with **higher attendance tend to score more marks**.
 3. Most students fall in **average performance category (Grade B and C)**.
 4. **Math subject shows more variation** in marks compared to other subjects.
 5. Overall class performance is **good with many students scoring above average**.
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Conclusion

This project shows how data analysis can help in understanding student performance. By using tools like **Pandas, NumPy, and Matplotlib**, we can easily analyze and visualize data.

The analysis helps teachers and students to:

- Identify strong and weak subjects
- Improve performance
- Focus on attendance

Overall, this project is useful for making **data-driven decisions in education**.